From Intra-Distinctiveness to Inter-Invariance: A Cycle-Resemblance Few-Shot Transformation Network for Cross-Domain Hyperspectral Image Classification

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***Abstract*— For large-scale mapping applications, cross-domain hyperspectral image classification (HSIC) has emerged as a highly promising research area. However, the classification accuracy decreased significantly when unseen classes emerged. Few shot learning (FSL) methods are adopted in cross-domain HSIC methods to address this problem. Despite this, existing cross-domain HSIC methods still have three key issues that hamper their classification capabilities: 1) previous works struggle to balance incorporating distinctive intradomain knowledge and managing model complexity in the face of significant domain representation differences; 2) previous works inadequately consider the limited capture capacity of interdomain intrinsic mutually invariant structures; and 3) previous works fail to capture the distinct characteristics of both head categories (e.g., urban buildings) and tail categories (e.g., urban corn) simultaneously when applying FSL to deal with unseen classes problem. In this article, we propose a cycle-resemblance few-shot transformation (CF-Trans) network to effectively handle the aforementioned challenges by integrating intradomain distinctiveness with interdomain invariance. To facilitate efficient intradomain feature aggregation for HSI, a novel lightweight intradomain attentive network is introduced. Different from previous works, to reduce the negative impact caused by inaccurate classifier predictions, from the perspective of interdomain knowledge transformation, a cycle-resemblance adversarial network is designed to capture the intrinsic mutually invariant structures. A dynamic label expansion mechanism is designed to capture the distinctive intradomain features of the head and tail classes. Experimental results on six HSI datasets including agricultural, rural–urban and urban datasets show the remarkably performance of our network.**

***Index Terms*— Cross domain, cycle-resemblance, few shot learning (FSL), hyperspectral image classification (HSIC), interinvariance, intradistinctiveness.**

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I. Introduction

HYPERSPECTRAL images (HSI) with abundant spectral information have gained widespread application in various fields [1], [2], [3], including urban mapping, environmental monitoring, agriculture, and mineral resources exploration. Hyperspectral image classification (HSIC) has been a prominent study area for remote sensing interpretation, focusing on assigning distinctive class labels to individual pixels [4], [5]. In recent years, there has been a broad application of machine learning [6], [7], [8] and deep learning approaches [9], [10], [11] for HSIC. Zhang et al. [12] proposed a spatial aggregation network leveraging morphological transformations to identify fine-grained boundaries in HSIC. Wang et al. [13] proposed a multicale interactive information extraction method to fusion HSI and synthetic aperture radar image. Zheng et al. [14] proposed a foreground-aware relation network for remote sensing object segmentation. Li et al. [15] proposed a graph-feature-enhanced selective assignment network for HSIC. While these methods have achieved good results in HSIC, these methods assume that the data are from the same data distribution [16]. Cross-domain refers to transferring knowledge or models from source domain (SD) to target domain (TD) to assist in classifying the TD. Direct transfer across significantly different data distributions can cause a notable decline in classification performance, limiting its applicability in practical scenarios [17]. The key challenge of HSIC cross-domain methods focuses on simultaneously capturing intradomain distinctiveness knowledge and interdomain invariant knowledge within a unified framework. Intradomain distinctiveness refers to the specific data characteristics within either the SD or TD, whereas interdomain invariance denotes shared characteristics between them.

Significant domain differences and label scarcity in the TD make acquiring intradomain distinctiveness challenging, yet comprehensive domain-specific knowledge acquisition enhances access to cross-domain invariant knowledge. As shown in Fig. 1(a), this involves three main challenges.

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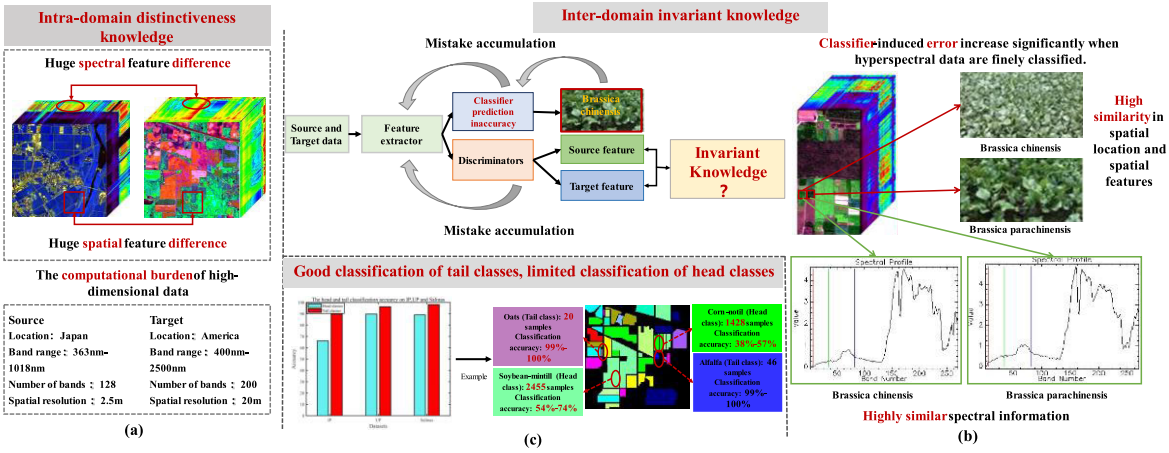


Fig. 1. Three main challenges. (a) Capturing effective intradomain distinctiveness knowledge becomes challenging in the presence of significant domain differences. (b) Insufficient consideration for classifier prediction inaccuracy in capturing interdomain invariant knowledge. (c) Performance of the head and tail categories cannot be balanced.

The first lies in the indiscriminate treatment of spatial information by the 3DCNN method, which introduces significant interference [18]. The second is HSI holding abundant spectral details, where each class exhibits its distinctive optimal spectral set [19]. The third is high-dimensional nature of HSI that also bring significantly computation [20]. Attention mechanisms weight spectral and spatial information highlight crucial spatial-spectral features [21]. When applied to cross-domain HSIC task, attention mechanisms enhance the perception of curial interdomain variant knowledge, thereby mitigating the impact of information discrepancy [22]. Huang et al. [23] constructed attention-based dual-branch feature extraction sub-network to carry out cross-domain HISC task. Mdrafi et al. [24] proposed attention transfer architecture for domain adaptation in HSIC. Fu et al. [20] devised two compact attention modules based on the self-attention mechanism to capture critical spatial and spectral features. CBAM is a straightforward and efficient attention module that can be easily integrated into CNN architectures [25]. Inspired by CBAM, Sun et al. [26] proposed the double-branch multiattention mechanism network (DBMA) to excavate the spectral and spatial information. Ma et al. [27] evaluated the effectiveness of combining CBAM with CNN for tree species classification in UAV-based HSI. Hyperspectral data typically comprises tens to hundreds of channels, wherein calculating the relationships between each pair of channels results in significant information redundancy. Currently, most CBAM-based approaches mitigate model complexity by integrating CBAM with lightweight networks [28], [29]. Yang and Jiang [30] employed dilated convolutions and integrated spatial and channel attention mechanisms to reduce computational complexity while efficiently extracting residual edge path features and enhancing image information embedding. The conventional sequential multiplication approach in CBAM may result in information loss, especially in cross-domain HSIC tasks characterized by significant feature variation. This could potentially compress or overlook critical details, while the substantial computational burden imposed by high-dimensional hyperspectral datasets is not adequately addressed. Thus, effectively integrating simple CBAMs with minimal model complexity to capture crucial spatial-spectral features in hyperspectral images while minimizing detail loss presents a significant challenge.

Given the substantial differences in data distributions between the two domains, the acquisition of domain-invariant knowledge has been demonstrated as a challenging task [16], [31]. Domain adaptation methods for cross-domain HSIC tasks can be divided into two main methodologies: discrepancy-driven domain adaptation methods [32], [33] and adversarial-inspired domain adaptation methods [34], [35]. Zhang et al. [33], [36] adopted maximum mean discrepancy (MMD) for marginal distribution alignment of cross-domain HISC task. Zhu et al. [37] proposed the concept of subdomains and used local MMD for subdomain alignment. However, the computation of MMD relines on adopting a fitting kernel mapping, and the choice of kernel mapping influences the results significantly [38], [39]. The current adversarial domain-based adaptation approach has achieved good results and has also been applied to cross-domain HSIC tasks. Ma et al. [40] proposed an unsupervised cross-dataset HSIC method with an adversarial domain-based adaptation, which utilizes multiclassifiers to construct discriminators and employs covariant autoencoders to constitute generators. Ma et al. [17] introduced an innovative metric for unsupervised domain adaptation in land-cover classification, emphasizing local consistency and global diversity. As shown in Fig. 1(a), the data distributions in SD and TD exhibit significant difference, which presents multimodal data distributions. The multimodal data structure can capture through the cross-entropy loss between classifier predictions and feature [41]. Zhang et al. [42] employed conditional domain discriminator to capture multimodal interdomain variant knowledge. This method achieved good classification results, but the predictions of the classifiers are uncertain. Although entropy is proposed to reduce the classifier prediction inaccuracy problem, it has own uncertainties. In Fig. 1(b), both Brassica parachinensls and Brassica chinensis belong to the Brassica category.

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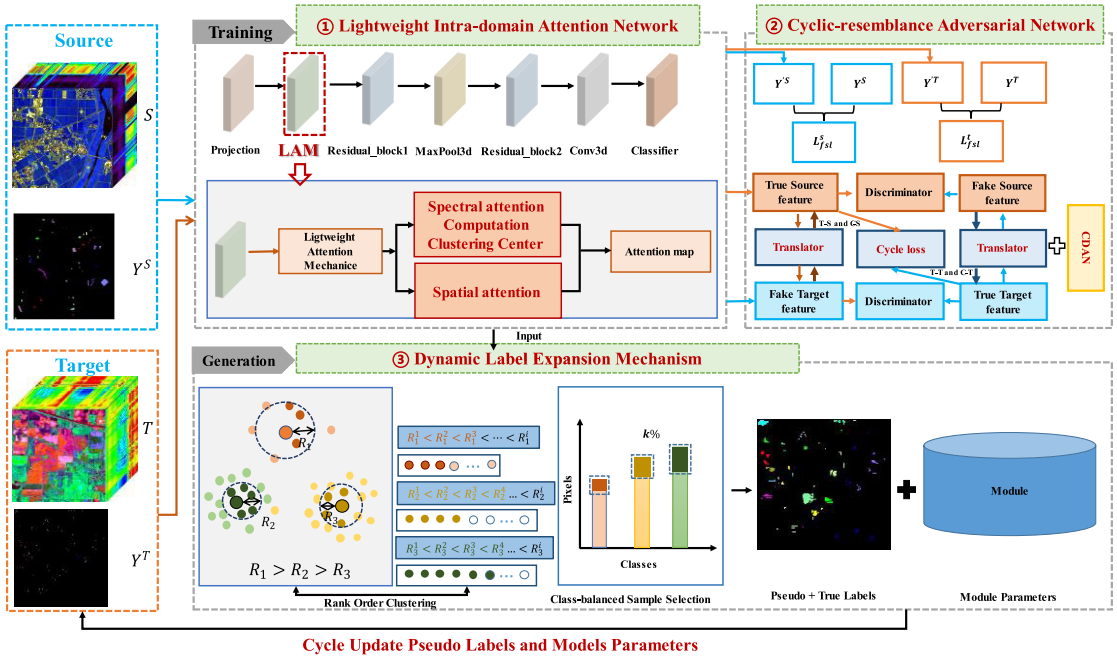


Fig. 2. Architecture of the proposed CF-Trans framework. T–S and C–S are the true SD feature and the SD feature after one cycle of transformation. T–T and C–S are the true TD feature and the TD feature after one cycle of transformation. The training and generation processes are alternatively automated.

They share highly similar spectral curve, textures, and are located very close to each other spatially. HSI is commonly utilized for fine classification tasks, and the classifier is more prone to misclassification in HSIC tasks than in other classification tasks. Therefore, the captured domain-invariant knowledge may exhibit substantial bias, even if the discriminator is completely confused [41], [43]. As shown in Fig. 1(b), based on the status of Brassica parachinensls being misidentified as Brassica chinensis on the discriminator, Brassica parachinensls will be permanently classified as soybean. In the worst case, the error caused by the prediction may be propagated and magnified. CycleGAN is employed in HSIC tasks for its capability to effectively learn domain-mapping relationships. Wang et al. [44] used CycleGAN to create a bi-directional mapping between SD and TD features, capturing interdomain invariant features. Effective feature mapping mitigates classifier prediction inaccuracies, yet the capture of multimodal data remains crucial. Therefore, how to design a network that mines intrinsic domain-invariant knowledge and multimodal data structure in the presence of inaccurate classifier predictions is important for cross-domain HSIC tasks.

Domain adaptation methods often suffer significant performance degradation when new classes emerge in the TD [45]. In recent years, few shot learning (FSL) algorithm has received much attention due to being able to identify new unseen classes using a small number of training samples [36], [45]. Head categories indicate more numerous, while tail categories indicate less [46], [47]. For FSL method, the head classes usually exhibits a very limited number of samples, and the model may not perfectly understand and trap the feature changes in the head classes, resulting in poor classification results [48]. As shown in Fig. 1(c), FSL method demonstrates strong performance in capturing intradomain distinctiveness knowledge to tail-classes (e.g., Oats and Alfalfa), while exhibiting limited efficacy in extracting knowledge related to head-classes (e.g., Soybean-mintill and Corn-notill) [48]. To comprehensively validate the inefficacy of FSL methods in head class classification, we statistically analyzed the classification accuracies of heads and tails for 10 FSL methods across IP, UP, and Salinas datasets. The experimental results indicate significantly better classification performance for the tail classes compared to the head classes. The problem of sample scarcity in head classes has persisted in the FSL method, but was never noticed and solved. The semisupervised approach in cross-domain frameworks offers a novel perspective for addressing the challenge of poor head classes classification performance, achieved through label expansion [49], [50]. Yang et al. [51] proposed a multiscale dynamic convolutional network based semisupervised HSIC framework. Fang et al. [9] proposed a semisupervised HSIC method employing deep clustering, which aims to cluster deep features and achieve labels expansion of unlabeled samples. Although semisupervised classification methods have shown effectiveness in HSIC tasks, there has been limited research on integrating the semisupervised HSIC methods to cross-domain framework to address the challenge of FSL focusing on mostly on the tail classes. Therefore, how to incorporate semisupervised learning in a cross-domain HSIC framework to enhance the HSIC power of head classes is important for cross-domain HSIC tasks.

To address the above problems, a cycle-resemblance few-shot transformation (CF-Trans) network is proposed which mainly consists of lightweight intradomain attention (LIDA) network, cyclic-resemblance adversarial network (CRAN) and dynamic label expansion mechanism (DLEM). CF-Trans network aimed at capturing both intradomain distinctiveness and interdomain invariant knowledge. The comprehensive

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acquisition of intradomain distinctive knowledge not only effectively focuses on the unique characteristics of a specific domain but also plays a crucial role in enhancing access to interdomain invariant knowledge. To capture intradomain distinctive knowledge while reducing computation and information loss, LIDA employed a parallel summation mechanism to focus on critical spatial-spectrum features. To accurately capture multimodal interdomain invariant knowledge and mitigate the impact of inaccurate classifier predictions, this article introduces a CRAN integrated with a conditional discriminator. The classification performance of cross-domain HSIC methods significantly degrades when encountering unseen categories. To preserve the advantages of the FSL method in addressing the unseen class problem while tackling the classification imbalance between head and tail classes, this article proposes DLEM.

The key contributions of this task are listed below.

1) To efficiently perform intradomain feature aggregation and reduce information loss while balancing efficiency and accuracy, the LIDA network is proposed for HSI feature extractor. To cope with the computational burden of high-dimensional features in HSI, the idea of aggregation centers proposed to present the representative spectral signal. To circumvent information loss due to substantial feature differences of HSI, a method involving parallel fusion and integration of low-level with high-level features has been introduced.
2) For capturing the multimodal data structure of HSI and reducing the impact of classifier prediction inaccuracies on fine HSIC tasks, the CRAN method based on interdomain knowledge mapping idea is introduced. CRAN consists of two knowledge translators and their associated sample discriminators, which facilitate interdomain knowledge transformation and integrate conditional domain adversarial networks to effectively capture multimodal domain invariant knowledge. Additionally, we introduce a cycle consistency loss term to ensure the preservation of original data characteristics throughout the transformation process.
3) To overcome the poor performance of FSL method in head-classes HSIC, the DLEM method, integrating the semisupervised ideology, has been devised. This mechanism involves ranking the clustering distances and applying a class-balanced sampling strategy to select and dynamically update pseudolabels. By ensuring the quality of the pseudolabels and enriching the labels of the head classes, this approach enables a more comprehensive capture of the feature variations within the head classes.

The subsequent sections of this article are structured as follows. Section II gives some cross-domain FSL and CNN-related concepts and literature review. In Section III, the CF-Trans network for cross-domain HSIC is elaborated. In Section IV, a description of the dataset and an interpretation of the experimental results are given. In the end, conclusions will be made in Section V.

II. Related Works

This work is based on FSL and CNN. Therefore, we briefly review these methods.

A. CNN for Cross-Domain HSIC

CNN use a set of parameters that refers to a kernel function or kernel-to-extract features [52], [53]. It has been widely used for HSI classification. CNNs have three main characteristics: local connectivity, the same convolutional kernel sharing the same parameters, and shift invariance. These three main features make CNNs have powerful feature-extraction capabilities. The convolutional layer contains three main components: a linear mapping, an activation function, and a pooling function [54]. Hu et al. [55] introduced a novel CNN method, but only used spectral information. In early HSIC tasks, 2-D convolution was mainly used, mainly for extracting spatial texture information. Zhang et al. [56] proposed 1-D CNN and 2-D CNN methods to individually pick up spectral and spatial features. For simultaneous usage of spectral-spatial information, SSRN [57] and DBMA [58] methods are devised. In this task, 3DCNN is taken as the primary feature extraction architecture. This approach can effectively capture spatial-spectral features and address the gradient disappearance issue using residual linkage.

B. FSL for Cross-Domain HSIC

In FSL, there is a term called "C-way K-shot" problem. The support set is generated by randomly selecting C categories, with each category consisting of K samples. The query set is formed by selecting samples from the remaining data within these categories. Essentially, the model learns how to recognize these classes from $C \times K$ data. Based on the principles of FSL method, it has been employed to address the challenge of encountering a new dataset with unseen classes [59], [60]. Li et al. [45] pioneered the integration of domain adaptation ideas and FSL in HSIC task. Xi et al. [61] introduced a HISC framework based on class-covariance measures which employ an interactive studying methodology. Zhang et al. [36] presented a cross-domain HSIC that utilizes graph convolution and FSL. Li et al. [62] introduce an asymmetric encoder–decoder framework for cross-domain FSL HSIC. Qin et al. [63] innovate in cross-domain FSL HSIC with their feature disentanglement approach.

The FSL method can handle unseen classes, but optimizing the classification performance for both head and tail classes is challenging when samples are extremely limited in the target domain. Therefore, several critical issues need to be addressed. How to balance efficiency and accuracy in digging more effective intradomain distinctiveness knowledge and reduce information loss? How to minimize classifier prediction inaccuracies in fine classification tasks while enhancing the capture of effective multimodal interdomain invariant knowledge? How to effectively improve head-class category classification within FSL method? To solve these problems, the cross-domain HSIC CF-Trans network is proposed.

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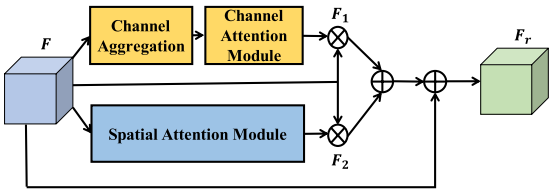


Fig. 3. Lightweight attention mechanism.

III. Proposed Method

Fig. 2 depicts the CF-Trans flowchart, which contains the extraction of intradomain distinctiveness knowledge and interdomain invariant knowledge. To begin with, the support and query sets are formed by leveraging the SD and TD datasets. The projection layer to guarantee the uniformity of feature dimensions at SD and TD. To capture distinctiveness knowledge within a domain effectively while minimizing computational and information loss, LIDA employs a parallel summation mechanism to prioritize crucial spatial-spectral features. We introduce a CRAN aimed at accurately capturing multimodal interdomain invariant knowledge and mitigating inaccuracies in classifier predictions. To address the significant degradation in classification performance when encountering unseen categories in TD, CF-Trans employs the FSL method. While FSL effectively tackles the unseen class problem, it contends with the imbalance between head and tail classes. To address this, this article introduces the DLEM.

A. LIDA Network

Different sensors for acquiring data lead to different dimensions in the SD and TD. The projection layer is introduced for ensuring that the input sample has the same dimension before through lightweight attention mechanism (LAM). Therefore, before feature extractor, the two projection layers P^S and P^T ensure that the SD (e.g., 128 bands) and the TD (e.g., 204 bands) have the same dimensions. The projection layer is made by a 2-D CNN. Here, $P \in R^{W\times H\times \mathrm{ch}}$ is the input of HSI, $W \times H$ represents the spatial dimension, ch denotes the band numbers. The projection layer yields an output feature expressed as $P' = P \times T (T \in R^{\mathrm{ch}\times 100}, P' \in R^{W\times H\times \mathrm{ch}})$. A 3-D residual CNN is employed to conduct feature extraction. FSL is simultaneously employed utilizing in the SD and TD. FSL takes place through calculating the distance among the unmarked samples and the marked samples for each category.

To efficiently perform intradomain feature aggregation and reduce information loss while balancing efficiency and accuracy, we propose an LAM. The full LAM flowchart is shown in Fig. 3 which contain spatial and spectral attention mechanism. Compared to CBAM, we designed feature clustering, parallel summation, and low-level and high-level feature fusion mechanisms to better focus on critical spatial and spectral features while reducing computational burden and minimizing information loss. Number of channels for hyperspectral data is in the tens to hundreds, for which calculating each channel-to-channel relationship leads to information redundancy [20]. To solve this problem, we compute the relationship between aggregation centers and aggregation centers. Spectral attention weights were obtained through the spectral attention mechanism. Then, the spectral attention weights were multiplied with the original features to obtain the feature weighting result F_1 to highlight efficient spectral features and inhibit interfering spectral features. Meanwhile, through multiplying the features processed by the spatial attention mechanism with the original features, a weighted feature map F_2 was obtained to focus on the effective spatial features and suppress the interfering spatial features. To achieve feature fusion, the weighted feature maps F_1 and F_2 are added together. For certain feature values that are small, multiplication may yield even lower values, thereby leading to information loss. This could cause smaller feature values to become further diminished, posing challenges in effective capture. Finally, to reduce the information loss, the fused features are combined with the low-level features so that the model considers both the original and the processed features in the overall performance, thus improving the comprehensive performance.

We define the clustering center as a condensed spectral map formed by aggregating the spectral information of the input features. First, a 1×1 convolutional layer is constructed to squeeze the spectral dimension of feature A to get feature F. The individual spectral maps within F are considered as clustering center. Subsequently, a lightweight spectral attention map is acquired by maximum pooling (MaxPool) and average pooling (AvgPool). The two output feature vectors are fused using a multilayer perceptron (MLP). Finally, by applying an activation function to the output feature vector, the spectral attention map ($M_c(F)$) is generated

$$M_c(F) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(\mathrm{Conv}(A)) + \mathrm{MLP}(\mathrm{MaxPool}(\mathrm{Conv}(A))). \quad (1)$$

In the HSI data cube, neighboring pixels contribute differently to the central pixel. The spatial attention mechanism directs its focus toward specific regions and performs a reweighting operation on the resulting attention regions. Two spatial feature mappings are acquired by employing an average pooling layer and a maximum pooling layer. The two feature maps are stitched together and then used the activation function and convolution operation to generate a 1-D feature map. The formula is

$$M_s(F) = \sigma(\mathrm{Conv}(\mathrm{AvgPool}(A); \mathrm{MaxPool}(A))). \quad (2)$$

B. Cyclic-Resemblance Adversarial Network

The SD and TD have abundant intradomain distinct knowledge and interdomain invariant knowledge. For example, in the spectral information of Fig. 4, SD and TD show similar trends in the vegetation spectral curves, demonstrating interdomain invariance. In contrast, the road spectral curves of the SD and TD show obvious differences, highlighting the intradomain distinctiveness. Regarding spatial information, the road morphology in two domains exhibits similarity, indicating interdomain invariant knowledge. However, the spatial arrangement of roads differs notably between the SD and TD, presenting intradomain distinctiveness. Meanwhile,

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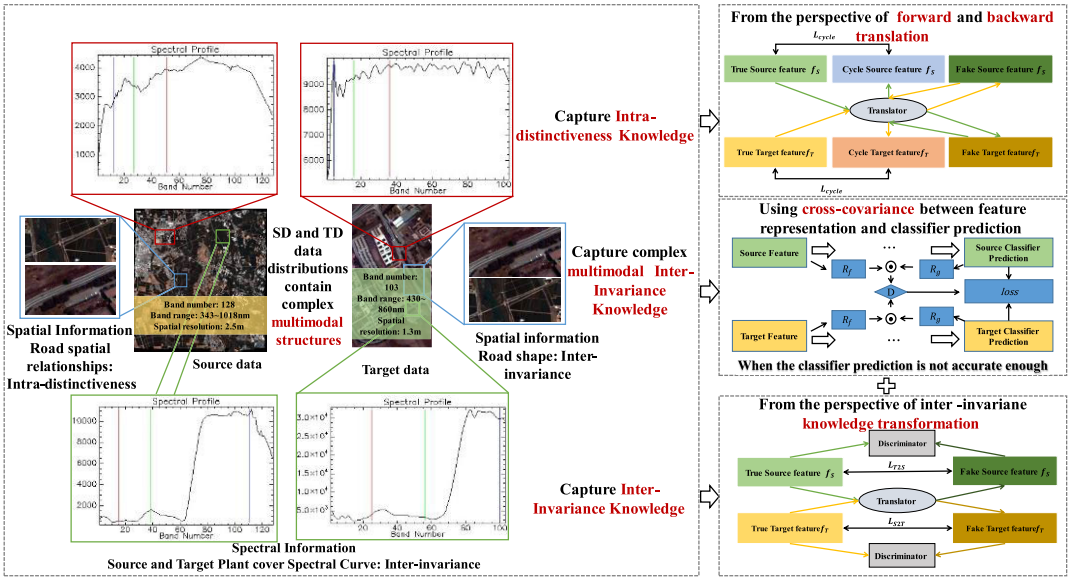


Fig. 4. CRAN. From the perspective of forward and backward translation to capture intradistinctiveness knowledge. From the perspective of interinvariant knowledge transformation to capture interinvariance knowledge. Applying cross-covariance between feature representation and classifier prediction to obtain complex multimodal interinvariance knowledge.

the substantial divergence between the two domains fosters a rich multimodal data structure. The multimodal data structure in cross-domain HSIC refers to the scenario where data acquired by different sensors lead to substantial differences between SD and TD in terms of the number of bands, band range, and spatial resolution. Multimodal data structure can be adequately captured by cross-covariance dependencies between features and classes. Multicollinear conditioning captures cross-covariance between feature representation and classifier prediction, entropy conditioning controls uncertainty in classifier prediction to ensure moveability. Therefore, multilinear conditioning and entropy conditioning are employed to capture multimodal data structures. However, multilinear conditioning is likely to cause dimensional explosion. Inspired by CDAN [43], random linear conditioning is used here instead of linear conditioning to solve the dimensional explosion. The formula of the algorithm to get domain invariant embedding knowledge is

$$\min_{D}\max_{f,g} L_{\text{con}} = -E_{x_i^s\sim p_s(x)}\log\left[d\left(f_i^s, g_i^s\right)\right] - E_{x_j^t\sim p_t(x)}\log\left[1-d\left(f_j^t, g_j^t\right)\right] \quad (3)$$

where d is conditional domain discriminator, g_i^s is source classifier prediction, g_i^t is source classifier prediction, f_i^s is source feature, f_i^t is target feature, $d(,)$ is the probability that the prediction is the SD. $1-d(,)$ is to predict the probability of owning the TD.

HSI is often utilized for fine feature classification task, for example, in distinguishing between different varieties of rice. This means that the classifier may make more prediction errors. When the classifier prediction is not accurate enough, misconfiguring the conditions for the discriminator can expose the feature extractor to a precarious state. Although CDAN has proposed the entropy condition, the entropy itself is inaccurate. Accurate interdomain invariant knowledge enables interdomain transformation [41]. The CRAN is focused on

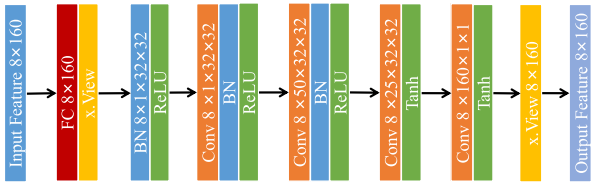


Fig. 5. Knowledge translator.

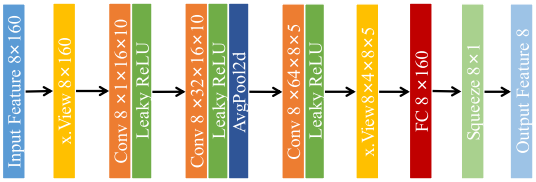


Fig. 6. Domain sample discriminator.

bidirectional mapping to establish the mapping relationship of interdomain features. Feature translators and sample discriminators are trained to enable efficient transformation of interdomain data. It helps to make the model more effective for domain adaptation when dealing with the larger differences between domains. Therefore, we train two knowledge translators in Fig. 5 to translate the knowledge among the SD and the TD. Two discriminators to distinguish real knowledge from translated knowledge are trained.

The knowledge translators T_{s2t} and T_{t2s} were evaluated in an adversarial manner using sample discriminators d_T and d_S in Fig. 6. A TD sample discriminator d_T is trained to distinguish between false TD knowledge from SD translations and true TD knowledge. A SD sample discriminator d_S is trained for identifying false between false SD knowledge from TD translations and true SD knowledge. Therefore, we define the following loss function:

$$\min_{T_{s2t}}\max_{d_t} l_{s2t} = E[\log d_t(f_t)] + E[\log(1-d_t(T_{s2t}(f_s))] \quad (4)$$

$$\min_{T_{t2s}}\max_{d_s} l_{t2s} = E[\log d_s(f_s)] + E[\log(1-d_s(T_{t2s}(f_t))] \quad (5)$$

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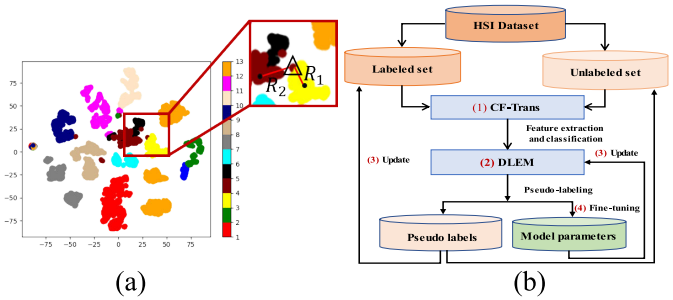


Fig. 7. DLEM. (a) Visualization of feature embeddings using 2-D t-SNE for a subset of 4959 randomly selected unlabeled samples from Kennedy Space Center dataset. R_1 and R_2 represent the distances from the dark red samples marked by triangles to the centers of two different clusters. (b) Overview of the whole classification process. The proposed framework includes 1) Model initialization. 2) Pseudo-labeling. 3) Labeled and unlabeled update. 4) Model fine and inference.

where f_s is source feature, f_t is target feature, l_{s2t} is from source to target loss, l_{t2s} is from target to source loss, T_{s2t} is knowledge translators from source to target feature translators, T_{t2s} is knowledge translators from target feature to source feature, d_S is source discriminators, and d_T is target discriminators.

Intradomain distinctiveness makes the same category exhibit very different characteristics. As shown in the road of Fig. 4, its spectral profile and spatial structure in both SD and TD show their respective distinctiveness. Intradomain distinctiveness can help the model to better distinguish different categories, thus improving the accuracy of classification. HSI contains rich spectral and spatial information, which provides rich information about intradomain details. To extract the interdomain invariant knowledge while preserving the rich intradomain detail information of HSI, the consistency loss (L_{cyc}) is applied to preserve the intradomain distinct knowledge

$$\min_{T_{t2s},T_{s2t}} L_{\text{cyc}} = E\left[\|T_{t2s}(T_{s2t}(f_s)) - f_s\|_2^2\right] + E\left[\|T_{s2t}(T_{t2s}(f_t)) - f_t\|_2^2\right]. \quad (6)$$

The CRAN is shown in Fig. 4 which consists of two knowledge translators along with their sample discriminators and conditional adversarial network. Thus, the overall loss function of the domain adaptation is

$$L_D = L_{\text{con}} + \beta_1(l_{s2t} + l_{t2s}) + \beta_2 L_{\text{cyc}} \quad (7)$$

where β_1 and β_2 are weighting parameter.

C. Dynamic Label Expansion Mechanism

By employing DLEM we can provide more training data to capture the unique features in the head classes, thus improving the HSIC performance in the head classes. In spatial distribution, those that are close to each other are more likely to belong to the same category. As shown in Fig. 7(a), the dark red points are closer to the yellow center sample and are classified as yellow category, but it actually belongs to the dark red category. The quality of pseudo labels can be enhanced by configuring a specific distance threshold. However, setting the distance threshold may result in certain categories having fewer or even no pseudo labels, as illustrated by the green category in Fig. 7(a). Inspired by the principle of class-balanced sample selection, we adopted different distance thresholds for individual classes to balance the head-tail classes sample numbers. Specifically, we rank the clustering distances for each class and applying class-balance sample selection dynamically adapted the thresholds of every class to get reliable pseudo-labels for each class. This whole HSIC process is shown in Fig. 7(b), where our framework continuously updates both labeled and unlabeled data through a dynamic label extension mechanism and fine-tunes the model parameters to achieve better classification results. DLEM effectively increases the number of head class samples, balancing their classification performance. Concurrently, the classification performance of the tail class remains robust due to its abundance of ground truth labels.

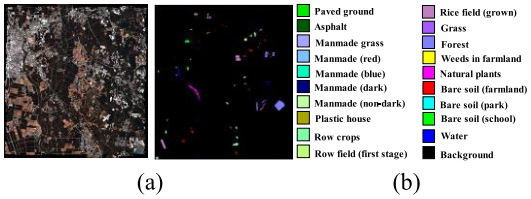


Fig. 8. Chikusei: (a) FC image and (b) GT.

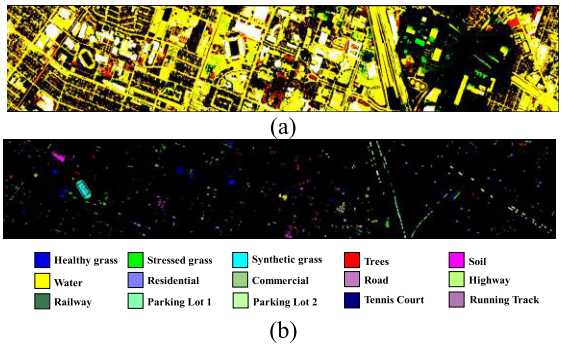


Fig. 9. HU: (a) FC image and (b) GT.

IV. Experiments Result and Analysis

In practical scenarios, SD and TD often exhibit disparities in terms of feature distributions and classes. Therefore, five diverse types of SD data are taken to confirm the validity of CF-Trans.

A. Experimental Datasets

1) Large-Scale and Multicategory SD Datasets: Chikusei: The Headwall Hyperspec-Vnir-C sensor was applied to capture this dataset. This comprises 2517 × 2335 pixels, 128 spectral bands extending 343–1018 nm, 19 categories, and has a spatial resolution of 2.5 m. Category information can be found in Table I. Fig. 8 presents the false color (FC) image and the related ground truth.

2) Target Data:

a) Large-scale high-resolution urban dataset: Houston University (HU). Over the University of Houston campus, the National Center for Airborne Laser Mapping took this data. This data includes 349 × 1905 pixels, 144 bands spanning from 0.36 to 1.05, and possesses a spatial resolution of 2.5 m.

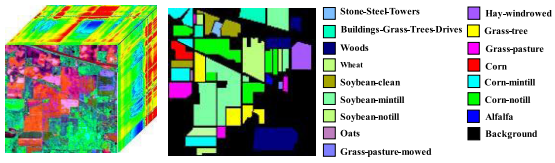
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TABLE I
CATEGORIES AND SAMPLE NUMBERS IN CHIKUSEI AND HU DATASETS

Chikusei-SD					HU-TD			
Class	Name	Total	Train	Test	Name	Total	Train	Test
1	Water	2845	200	2645	Healthy Grass	1251	5	1246
2	Bare soil(school)	2859	200	2659	Stressed Grass	1254	5	1249
3	Bare soil	286	200	86	Synthetic Grass	697	5	692
4	Bare soil(farmland)	48525	200	48325	Tree	1244	5	1239
5	Natural plants	4297	200	4097	Soil	1252	5	1247
6	Weeds in farmland	1108	200	908	Water	325	5	320
7	Forest	20516	200	20316	Residential	1268	5	1263
8	Grass	6515	200	6315	Commercial	1244	5	1239
9	Rice field(grown)	13369	200	13168	Road	1252	5	1247
10	Rice field(first stage)	1268	200	1068	Highway	1227	5	1222
11	Row crops	5961	200	5761	Railway	1235	5	1230
12	Plastic house	2193	200	1993	Parking Lot 1	1234	5	1229
13	Manmade(non-dark)	1220	200	1020	Parking Lot 2	469	5	464
14	Manmade(dark)	7664	200	7464	Tennis Court	428	5	423
15	Manmade(blue)	431	200	231	Running Track	660	5	655
16	Manmade	222	200	22	---	---	---	---
17	Manmade grass	1040	200	840	---	---	---	---
18	Asphalt	801	200	601	---	---	---	---
19	Paved ground	145	---	---	---	---	---	---
	Total	77592	3600	73847	Total	15011	75	14936

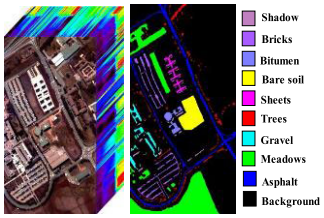
TABLE II
CATEGORIES AND SAMPLE NUMBERS IN IP AND UP DATASETS

IP-TD					UP-TD			
Class	Name	Total	Train	Test	Name	Total	Train	Test
1	Alfalfa	46	5	41	Asphalt	6631	5	6626
2	Corn-notill	1428	5	1423	Meadows	18649	5	18644
3	Corn-mintill	830	5	825	Gravel	2099	5	2094
4	Corn	237	5	232	Trees	3064	5	3059
5	Grass-pasture	483	5	478	Sheets	1345	5	1340
6	Grass-tree	730	5	725	Bare soil	5029	5	5024
7	Grass-pasture-mowed	28	5	23	Bitumen	1330	5	1325
8	Hay-windrowed	478	5	473	Bricks	3682	5	3677
9	Oats	20	5	15	Shadow	947	5	942
10	Soybean-notil	972	5	967	---	---	---	---
11	Soybean-mintill	2455	5	2450	---	---	---	---
12	Soybean-clean	593	5	588	---	---	---	---
13	Wheat	205	5	200	---	---	---	---
14	Woods	1265	5	1260	---	---	---	---
15	Buildings-Grass-Trees-Drives	386	5	381	---	---	---	---
16	Stone-Steel-Towers	93	5	88	---	---	---	---
	Total	10249	80	10169	Total	42776	45	42731



(a) (b)

Fig. 10. IP: (a) FC image and (b) GT.



(a) (b)

Fig. 11. UP: (a) FC image and (b) GT.

Table I gives the details of the 13 categories. The FC image and corresponding GT can be seen in Fig. 9.

b) Class extremely unbalanced and feature-highly similar low-resolution agricultural dataset: Indian Pines (IP). In 1992 from the Northwest Indian, AVIRIS collected this dataset with 200 bands ranging 400–2500 nm, spatial resolution of 20 m, and 145 × 145 pixels. A comprehensive breakdown of the 16 vegetation classes is listed in Table II. Fig. 10 depicts the FC image and its correlated GT.

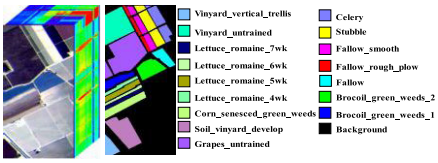
c) Class unbalanced high-resolution urban dataset: Pavia University (UP). The Reflective Optical Spectral Imaging System (ROSIS) was applied to collect this data. This is made up of 610 × 340 pixels, 103 bands covering 430–860 nm, and boasts a spatial resolution of 1.3 m. The detailed categories information is presented in Table II. The FC image and GT can be displayed in Fig. 11.

d) Class unbalanced and feature-highly similar high-resolution agricultural dataset: Salinas. In Salinas Valley, CA, USA, the AVIRIS sensor was employed to acquire this dataset. The data comprise 16 classes with images of size

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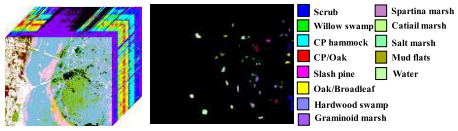
TABLE III
CATEGORIES AND SAMPLE NUMBERS IN SALINAS AND KSC DATASETS

Salinas-TD					KSC-TD			
Class	Name	Total	Train	Test	Name	Total	Train	Test
1	Brocoil_green_weeds_1	2009	5	2004	Scrub	761	5	754
2	Brocoil_green_weeds_2	3726	5	3721	Willow swamp	243	5	238
3	Fallow	1976	5	1971	CP hammock	256	5	251
4	Fallow_rough_plow	1394	5	1389	CP/Oak	252	5	247
5	Fallow_smooth	2678	5	2673	Slash pine	161	5	156
6	Stubble	3959	5	3954	Oak/broadleaf hammock	229	5	224
7	Celery	3579	5	3574	Hardwood swamb	105	5	100
8	Grapes_untrained	11271	5	11266	Graminoid marsh	431	5	426
9	Soil_vinyard_develop	6203	5	6198	Spartina marsh	520	5	515
10	Corn_senesced_green_weeds	3278	5	3273	Cattail marsh	404	5	399
11	Lettuce_romaine_4wk	1068	5	1063	Salt marsh	419	5	414
12	Lettuce_romaine_5wk	1927	5	1922	Mud flats	503	5	498
13	Lettuce_romaine_6wk	916	5	911	Water	927	5	922
14	Lettuce_romaine_7wk	1070	5	1065	---	---	---	---
15	Vinyard_untrained	7268	5	7263	---	---	---	---
16	Vinyard_vertical_trellis	1807	5	1802	---	---	---	---
	Total	54129	80	54049	Total	4959	65	4894



(a) (b)

Fig. 12. Salinas: (a) False-color image and (b) GT.



(a) (b)

Fig. 13. KSC: (a) FC image and (b) GT.

512 × 217 pixels, recorded 204 spectral bands spanning 400–2500 nm. The spatial resolution measures 3.7 m. Detailed categories information is provided in Table III. The FC image and GT presented in Fig. 12.

e) Huge class differences rural–urban dataset: Kennedy Space Center (KSC) Dataset. In Florida, on March 23, 1996, the AVIRIS sensor is employed to capture this data. It contains 13 categories, 512 × 614 pixels, 18 m spatial resolution, and 176 bands covering 400–2500 nm range. Further category information is summarized on Table III. Fig. 13 presents the FC image and GT.

B. Implementation Details

The Adam optimizer and Xavier normalization is applied to initialize all convolutional and linear layers. The SD applies 5 labeled samples of per class. At each training, query set is set to 19 samples per class, while support set is set to 1 sample per class. The training iteration count was configured to be 20 000. All experiments performed ten times. The learning rate is assigned to 0.001. The weight of knowledge translator β_1 is 0.00005 and the weight for consistency loss β_2 is 0.0005. The pseudo label cycle count is set to 5. The pseudo label selection number is 5%. The performance of the CF-Trans is evaluated using overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

C. Discussions About Classification Performance

For demonstrating that CF-Trans is effective, a visual and quantitative comparison was carried out with eight state-of-the-art HSIC approaches, which include traditional method (SVM-BRF [64]), CNN-based methods (3DCNN [65], DBMA [58], and SSRN [58]), global-based method (FPGA [66]), heterogeneous data parameter conversion method (HFSL [7]), and cross-domain methods (Gia-CFSL [36] and DCFSL [45]). To guarantee the fairness of the experiment, five labels are provided for each category in TD. For all cross-domain methods selected, there were 200 labels in SD for each category. CF-Trans is executed using the PyTorch framework platform.

The visual classification results for SVM-BRF, 3DCNN, DBMA, SSRN, FPGA, HFSL, Gia-CFSL, and DCFSL classification are given in Figs. 14(b)–(j) and 17(b)–(j). The CNN-based classification method is better than SVM in visualization and the images are smoother. This is attributed to the integration of spatial information features in CNN approach. FGPA methods have better classification performance compared to CNN-based methods, mainly owing to the fact that global-based methods can fully utilize the global information. HFSL methods have better visualization results than CNN-based methods, since FSL can extract features from fewer labeled datasets. Cross-domain methods were obtained as a land cover structure that is closer to the real image than the HFSL method. The visualization results show that the cross-domain approach can use the SD dataset to help TD classification. LAM can extract more discriminative knowledge. CF-Trans shows stronger classification performance for classes characterized by highly similar features, such as corn and soybean. The spatial-spectral resolution and data distribution of the IP dataset are very different from the SD, and CF-Trans fetches the best visualization results, showing that CRAN captured more accurate interdomain invariant knowledge. Based on the

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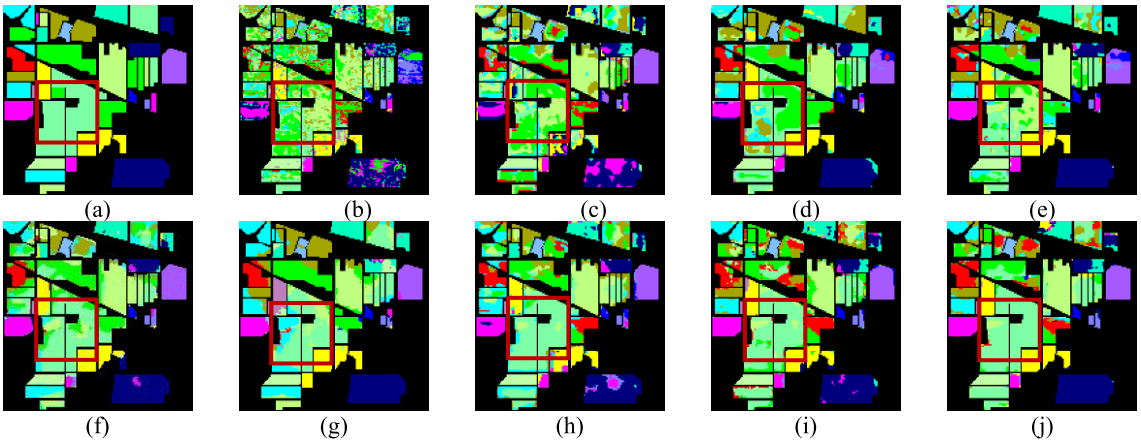


Fig. 14. Virtualization of classification graphs on IP datasets. (a) GT image, (b) SVM-BRF (44.96 ± 4.15%), (c) 3DCNN (45.4 ± 6.81%), (d) DBMA (57.81 ± 4.50%), (e) SSRN (64.91 ± 7.36%), (f) FGPA (69.79 ± 0.06%), (g) HFSL (69.19 ± 3.32%), (h) Gia-CFSL (63.92 ± 3.03%), (i) DCFSL (63.76 ± 4.19%), and (j) CF-Trans (73.98 ± 2.67%).

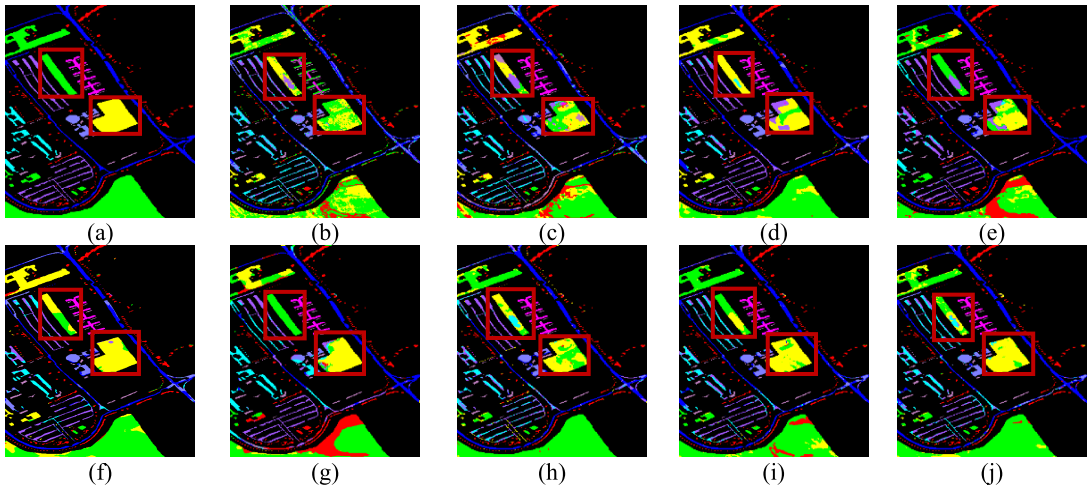


Fig. 15. Virtualization of classification graphs on UP dataset. (a) GT image, (b) SVM-BRF (58.70 ± 6.22%), (c) 3DCNN (62.12 ± 8.19%), (d) DBMA (70.91 ± 7.15%), (e) SSRN (73.37 ± 3.50%), (f) FGPA (75.99 ± 0.06), (g) HFSL (87.99 ± 2.84%), (h) Gia-CFSL (82.64 ± 3.30%), (i) DCFSL (83.73 ± 2.94%), and (j) CF-Trans (88.51 ± 1.98%).

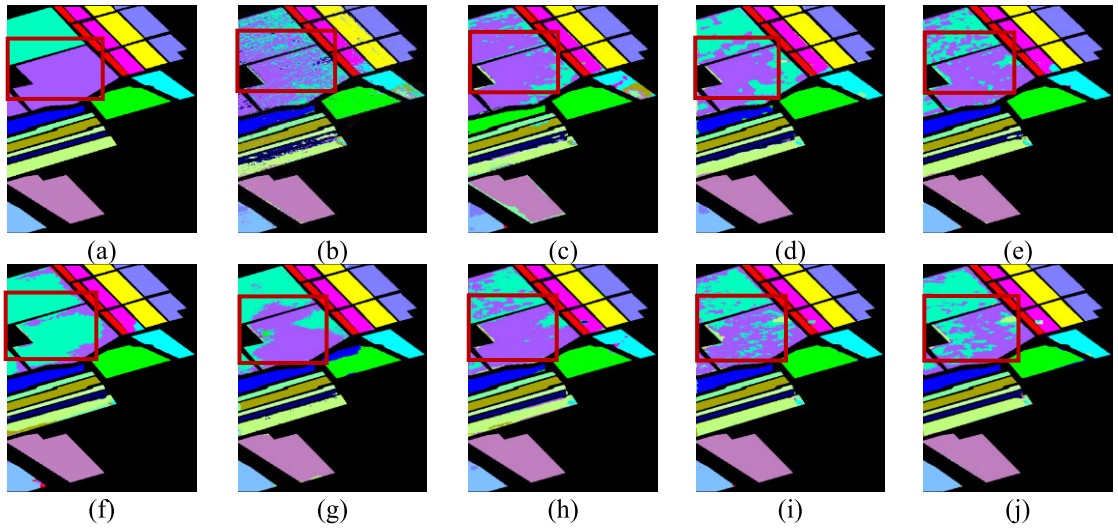


Fig. 16. Virtualization of classification graphs on Salinas dataset. (a) GT image, (b) SVM-BRF (80.29 ± 2.16%), (c) 3DCNN (69.01 ± 6.19%). (d) DBMA (87.08 ± 3.27%), (e) SSRN (88.14 ± 2.54%), (f) FPGA (80.52 ± 0.03%), (g) HFSL (88.50 ± 1.83%), (h) Gia-CFSL (89.04 ± 1.06%), (i) DCFSL (89.28 ± 1.89%), and (j) CF-Trans (91.03 ± 1.25%).

classification outcomes achieved from the Salinas dataset, although the classification accuracy of tail classes remains high, certain head classes, such as Grapes_untrained, exhibit significantly better performance compared to other methods. This indicates that the DLEM method can achieve better classification performance of head classes.

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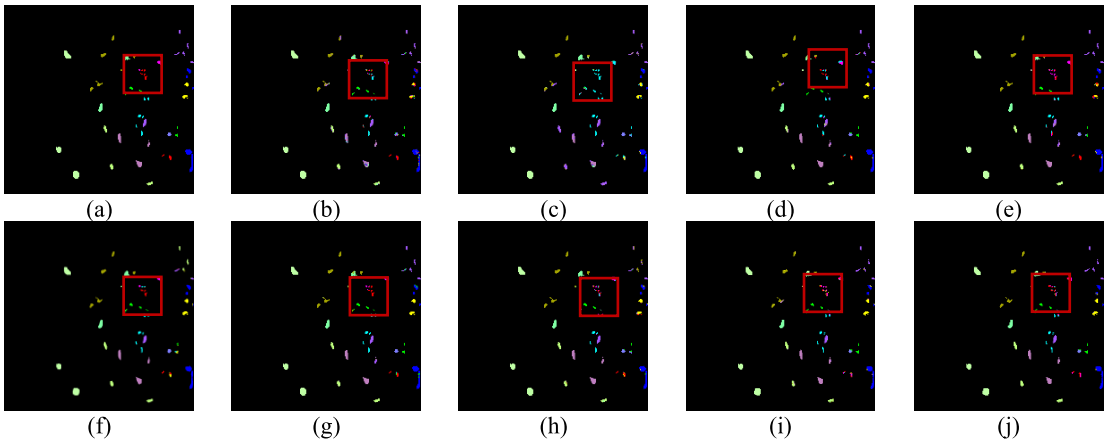


Fig. 17. Virtualization of classification graphs on KSC dataset. (a) GT image, (b) SVM-BRF (76.13 ± 3.97%), (c) 3DCNN (69.69 ± 4.72%), (d) DBMA (81.78 ± 3.21%), (e) SSRN (88.37 ± 1.22%), (f) FGPA (90.81 ± 1.57%), (g) HFSL (94.63 ± 1.70%), (h) Gia-CFSL (91.27 ± 1.35%), (i) DCFSL (92.04 ± 1.26%), and (j) CF-Trans (94.73 ± 0.38%).

TABLE IV

IP DATASET CLASSIFICATION RESULTS (%) (FIVE LABELED SAMPLES FOR EACH CLASS)

Class	SVM-BRF	3DCNN	DBMA	SSRN	FGPA	HFSL	Gia-CFSL	DCFSL	CF-Trans
1	19.44±5.17	29.96±10.11	32.57±17.17	43.91±25.75	92.68±0.00	**99.27±1.12**	85.61±18.81	92.38±10.85	95.12±11.59
2	38.89±6.06	40.04±16.32	53.18±10.44	**66.68±9.83**	50.95±0.07	47.28±12.98	43.44±10.21	43.37±12.10	61.56±7.73
3	25.75±10.11	21.68±5.57	39.56±17.21	47.28±21.06	54.18±0.12	**62.86±12.05**	48.70±9.10	47.23±12.74	62.01±10.09
4	17.33±4.05	26.22±9.78	41.15±11.84	45.69±17.75	74.35±0.22	**87.84±11.84**	77.28±14.44	70.39±17.56	87.37±6.61
5	48.65±9.13	49.02±15.06	77.26±19.16	87.49±9.67	73.74±0.10	73.39±10.18	72.87±7.24	72.05±7.19	75.06±6.11
6	83.24±2.97	70.27±8.02	89.52±7.08	**96.33±3.04**	87.45±0.00	80.41±13.23	83.20±6.24	85.44±5.83	91.57±5.25
7	21.36±13.39	18.69±11.63	19.73±7.82	26.23±11.01	**100.00±0.00**	99.57±1.30	99.57±1.30	99.58±1.25	99.57±1.31
8	89.91±3.53	89.58±8.30	99.95±0.16	97.11±7.71	**100.00±0.00**	96.49±6.10	90.32±4.68	82.62±11.98	90.63±6.99
9	11.77±3.02	11.83±4.05	5.09±1.96	25.15±16.68	**100.00±0.00**	**100.00±0.00**	99.33±2.00	99.38±1.88	**100.00±0.00**
10	35.1±7.96	40.01±8.23	49.74±14.63	53.2±15.54	**72.08±0.21**	56.08±14.13	61.32±7.53	55.59±12.43	67.52±6.07
11	54.23±9.02	55.35±11.78	64.48±6.97	**73.51±4.81**	60.27±0.14	61.31±15.75	60.84±17.24	59.91±16.02	67.32±10.58
12	20.33±4.86	22.6±10.15	33.16±6.35	41.66±12.90	50.68±0.00	**67.72±16.92**	43.88±13.15	40.46±9.19	58.66±15.81
13	77.72±6.06	57.64±11.09	90.47±7.97	96.27±5.59	99.50±0.00	98.10±2.33	98.75±1.40	99.25±0.68	**99.45±0.61**
14	82.41±4.62	70.4±5.86	93.67±5.47	**96.54±2.60**	88.21±0.04	87.79±13.04	76.99±13.78	86.03±6.77	93.18±5.79
15	35.11±6.24	38.7±9.16	60.28±12.95	69.67±9.92	**90.94±0.13**	89.84±10.04	66.72±13.19	70.60±11.34	78.32±11.79
16	84.5±25.22	56.17±12.57	80.55±12.73	84.55±7.25	**100.00±0.00**	96.70±4.78	97.39±4.88	98.20±2.98	98.07±3.90
OA	44.96±4.15	45.4±6.81	57.81±4.50	64.91±7.36	69.79±0.06	69.19±3.32	63.92±3.03	63.76±4.19	**73.98±2.67**
AA	46.61±2.97	43.63±4.50	58.15±3.03	65.71±3.61	80.94±0.01	81.54±1.76	75.39±1.94	75.16±1.69	**82.84±1.49**
Kappa	38.65±4.28	38.9±6.76	52.96±4.68	60.92±7.69	65.73±0.01	65.39±3.50	59.27±3.05	59.09±4.46	**70.55±2.93**

TABLE V

UP DATASET CLASSIFICATION RESULTS (%) (FIVE LABELED SAMPLES FOR EACH CLASS)

Class	SVM-BRF	3DCNN	DBMA	SSRN	FPGL	HFSL	Gia-CFSL	DCFSL	CF-Trans
1	90.79±9.24	91.17±5.04	**92.93±1.61**	99.14±3.56	78.31±0.05	75.71±9.00	82.42±6.09	80.38±9.47	86.94±6.30
2	78.73±6.56	84.81±4.52	89.28±4.74	**98.28±3.93**	66.28±0.15	90.39±6.29	86.13±7.27	87.95±5.92	93.53±3.45
3	34.22±9.40	29.72±9.92	56.99±18.91	61.27±12.71	58.99±0.05	**78.12±18.47**	66.54±6.19	63.87±9.50	62.84±15.08
4	55.24±15.93	61.28±25.11	80.95±20.21	51.81±20.56	92.55±0.07	92.07±7.46	95.77±2.10	95.25±2.33	**96.26±1.05**
5	90.26±7.74	86.16±10.65	98.57±1.59	99.82±2.23	99.14±0.04	99.92±0.14	98.35±1.61	99.46±0.96	99.53±0.39
6	31.28±9.96	43.56±10.76	42.79±14.00	63.25±11.04	95.52±0.10	**85.18±14.56**	73.22±9.47	78.44±6.99	82.76±9.42
7	37.94±7.55	35.45±12.38	50.15±17.37	68.17±8.15	81.77±0.11	**92.00±10.07**	78.85±7.10	80.35±7.47	79.72±8.42
8	65.82±9.21	61.01±6.07	68.32±5.73	60.82±9.99	73.72±0.01	**95.28±3.35**	67.90±11.37	68.97±9.47	78.19±11.37
9	99.77±0.30	46.16±22.02	89.35±7.85	97.92±0.93	99.84±0.05	99.50±0.90	98.41±2.06	98.84±1.89	**99.83±0.19**
OA	58.70±6.22	62.12±8.19	70.91±7.15	73.37±3.50	75.99±0.06	87.99±2.84	82.64±3.30	83.73±2.94	**88.51±1.98**
AA	64.89±2.83	59.98±4.98	74.37±4.26	78.63±3.11	82.90±0.01	89.80±2.87	83.07±1.00	83.72±2.00	**86.62±2.24**
Kappa	49.18±6.42	53.06±7.89	63.79±7.58	77.83±3.94	70.09±0.06	84.32±3.55	77.50±3.95	78.88±3.52	**84.90±2.53**

TABLE VI

SALINAS DATASET CLASSIFICATION RESULTS (%) (FIVE LABELED SAMPLES FOR EACH CLASS)

Class	SVM-BRF	3DCNN	DBMA	SSRN	FGPA	HFSL	Gia-CFSL	DCFSL	CF-Trans
1	97.87±3.48	37.45±19.28	96.98±8.87	97.88±3.86	**100±0.00**	98.67±2.11	98.97±1.07	99.13±1.13	99.08±0.95
2	98.75±0.51	52.06±35.54	99.33±0.85	98.86±2.12	**100±0.00**	95.26±7.59	98.702±1.71	99.83±0.22	99.81±0.26
3	81.52±6.98	80.62±10.76	94.78±5.45	93.12±3.41	**100±0.00**	92.62±8.18	91.98±8.61	92.19±10.60	95.38±4.95
4	97.31±0.43	78.94±14.10	89.82±3.44	97.85±1.04	99.10±0.03	99.46±0.76	99.16±0.79	**99.54±0.51**	99.44±0.50
5	95.55±3.53	91.75±5.73	**96.65±4.36**	93.82±6.78	93.42±0.03	94.99±6.36	90.13±3.04	92.45±3.05	92.32±1.95
6	99.63±0.56	98.25±1.26	99.31±0.80	99.98±0.02	98.36±0.00	99.36±0.76	99.36±0.79	99.00±0.97	99.13±1.14
7	94.51±3.12	92.00±3.28	96.76±3.64	99.56±0.53	**99.97±0.00**	93.75±15.82	98.31±1.99	99.33±0.91	99.47±0.68
8	68.83±3.16	64.78±12.27	79.53±8.19	74.6±6.15	27.00±0.00	71.70±9.80	77.56±5.50	76.20±12.59	**80.06±6.70**
9	98.19±1.97	91.92±3.73	98.59±0.71	99.29±0.33	**100.00±0.00**	98.76±1.45	98.60±1.31	99.42±0.94	98.97±1.16
10	79.67±8.26	87.26±7.04	89.03±5.05	**93.64±2.87**	68.62±0.25	90.54±8.25	84.26±6.10	84.60±9.00	87.31±5.30
11	75.51±8.47	47.00±18.91	83.84±14.29	93.09±3.48	**100.00±0.00**	99.45±0.43	96.56±2.90	97.71±3.04	98.53±1.14
12	86.06±9.10	76.21±7.70	98.93±1.36	97.78±1.04	99.17±0.00	98.56±1.16	99.63±0.30	99.25±0.34	**99.76±0.36**
13	84.52±14.02	66.80±19.72	81.88±12.38	95.33±6.78	**100.00±0.00**	98.14±2.45	98.52±1.95	99.25±0.61	98.78±1.02
14	77.51±16.14	75.50±26.36	88.29±7.62	95.48±0.02	**99.62±0.00**	96.25±3.10	97.84±2.01	98.72±1.05	98.82±0.78
15	51.63±6.87	44.6±20.55	63.96±10.86	68.05±0.53	87.82±0.11	76.97±13.57	73.54±5.33	73.47±7.44	**79.18±6.61**
16	80.02±20.55	80.38±17.99	95.48±4.91	99.77±0.37	93.54±0.02	92.19±8.51	88.64±5.06	91.81±5.83	91.27±4.58
OA	80.29±2.16	69.01±6.19	87.08±3.27	88.14±2.54	80.52±0.03	88.50±1.83	89.04±1.06	89.28±1.89	**91.03±1.25**
AA	85.44±1.54	72.84±6.02	90.82±2.45	93.63±0.68	91.66±0.03	93.45±1.26	93.23±0.87	93.89±1.02	**94.74±0.69**
Kappa	78.18±2.32	65.74±6.61	85.65±3.63	86.77±2.82	78.57±0.04	87.24±2.03	87.81±1.17	88.09±2.06	**90.03±1.38**

As shown in Tables IV–VIII, the CF-Trans network achieved higher HSIC accuracy over eight state-of-the-art methods on five different style TD datasets. The OA is improved by 1.75%–10.22%, AA is increased by 0.85%–7.68%, and the kappa accuracy is enhanced by 1.94%–11.46% for the cross-domain approaches. Owing to

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TABLE VII
KSC DATASET CLASSIFICATION RESULTS (%) (FIVE LABELED SAMPLES FOR EACH CLASS)

Class	SVM-BRF	3DCNN	DBMA	SSRN	FGPA	HFSL	Gia-CFSL	DCFSL	CF-Trans
1	93.22±3.14	93.55±6.58	99.18±1.63	91.88±6.74	**99.40±0.59**	94.64±4.39	95.95±4.17	96.85±1.63	97.42±1.69
2	79.82±7.45	46.85±32.56	89.38±12.65	**95.08±3.47**	73.74±1.47	85.04±15.15	81.89±12.46	85.63±8.52	90.46±9.12
3	66.28±8.03	35.68±15.42	63.63±26.86	89.60±9.23	**97.81±1.79**	95.50±5.14	82.63±12.06	85.42±11.53	93.27±9.41
4	38.43±10.14	28.09±13.81	56.76±19.92	68.04±19.52	68.83±7.29	**84.21±15.08**	52.23±12.50	51.94±10.28	61.50±13.41
5	54.48±13.63	16.31±8.64	53.28±27.29	62.77±18.69	72.12±12.5	**91.28±8.91**	82.12±5.82	73.59±10.23	78.08±8.69
6	42.01±10.60	53.64±13.65	61.13±25.07	65.84±14.20	89.73±10.27	94.55±13.32	85.76±8.63	92.59±5.58	**94.60±4.22**
7	69.98±5.77	48.61±25.79	63.49±12.42	84.23±6.99	**100.00±0.00**	**100.00±0.00**	96.50±5.80	94.00±9.54	97.20±3.66
8	59.07±13.50	42.48±19.13	68.05±7.50	83.04±8.96	53.76±3.99	89.34±14.76	87.49±7.81	88.22±8.58	**94.53±6.23**
9	69.88±5.62	73.94±12.11	88.22±11.79	87.65±8.95	90.10±9.90	92.47±6.92	92.43±6.60	93.75±8.78	**97.34±4.44**
10	82.44±14.74	50.14±20.29	86.71±13.05	96.46±5.49	**100.00±0.00**	97.84±4.69	97.69±2.33	97.94±1.82	98.85±2.40
11	90.07±5.12	89.27±7.58	94.95±7.87	96.07±5.39	**100.00±0.00**	98.02±1.93	95.12±4.38	95.56±4.48	98.24±1.58
12	82.41±7.66	91.01±10.16	83.40±11.45	94.67±3.98	**98.19±1.61**	95.40±5.69	94.22±5.01	95.38±2.32	96.29±2.85
13	99.02±1.92	96.67±2.50	95.31±7.65	98.78±2.44	**100.00±0.00**	99.99±0.03	**100.00±0.00**	99.97±0.10	99.98±0.07
OA	76.13±3.97	69.69±4.72	81.78±3.21	88.37±1.22	90.81±1.57	94.63±1.70	91.27±1.35	92.04±1.26	**94.73±0.38**
AA	71.32±3.69	58.94±5.31	77.19±3.93	85.70±1.07	87.97±1.93	93.71±2.29	88.00±1.32	88.53±1.55	**92.14±0.90**
Kappa	73.48±4.36	66.28±5.27	79.70±3.59	87.03±1.38	89.76±1.75	94.02±1.89	90.29±1.49	91.14±1.40	**94.14±0.42**

TABLE VIII
HU DATASET CLASSIFICATION RESULTS (%) (FIVE LABELED SAMPLES FOR EACH CLASS)

Class	SVM-BRF	3DCNN	DBMA	SSRN	HFSL	FPGA	Gia-CFSL	DCFSL	CF-Trans
1	82.02±5.80	60.53±3.00	80.08±9.63	83.2±13.53	77.86±11.71	95.66±3.33	89.24±6.75	89.13±6.70	**98.66±0.84**
2	84.48±11.70	51.79±32.44	**86.78±13.22**	83.47±15.54	57.65±17.48	75.18±4.38	83.07±7.46	83.25±9.74	84.07±0.90
3	74.75±13.70	32.00±17.59	97.94±4.94	91.45±15.60	**97.99±3.98**	95.05±6.32	96.16±3.17	97.12±1.95	96.05±2.07
4	90.11±13.35	68.73±4.64	86.61±12.58	75.03±38.65	85.10±6.28	**93.17±1.77**	90.20±3.85	92.11±2.10	91.71±0.22
5	89.69±3.34	78.25±24.47	94.74±4.71	88.58±12.07	83.11±17.98	**100.00±0.00**	96.50±2.04	97.09±3.77	94.77±2.21
6	87.63±15.91	22.55±24.41	94.62±6.82	**96.69±3.77**	89.53±4.08	90.99±3.11	77.72±7.44	80.72±6.18	88.02±4.14
7	57.69±9.66	45.6±21.26	69.55±10.23	68.56±7.73	81.30±10.37	69.94±5.64	63.01±9.51	70.13±8.94	**84.64±1.74**
8	68.56±11.86	45.64±21.90	**88.53±13.10**	83.25±16.50	41.72±6.31	53.86±6.42	40.78±10.84	41.92±7.88	40.68±6.21
9	58.02±8.02	40.46±23.76	74.37±11.80	58.33±16.35	78.00±12.40	**84.13±4.09**	58.81±12.01	64.15±5.32	79.95±2.94
10	50.11±5.96	7.5±10.73	58.52±12.15	48.77±26.12	72.34±17.89	**77.09±1.42**	64.77±17.19	63.23±17.01	67.29±13.26
11	46.18±8.07	38.49±34.55	68.56±13.26	55.99±11.83	**79.08±8.79**	55.23±7.70	61.41±12.72	62.15±7.59	72.52±5.26
12	53.25±8.28	18.73±13.76	69.98±10.93	66.14±9.87	64.50±12.42	45.28±3.38	68.12±11.96	74.42±12.65	**78.77±7.12**
13	18.76±9.38	60.31±9.72	64.58±12.77	68.04±15.87	92.44±4.36	64.14±4.41	88.10±4.11	88.04±5.70	**96.12±1.70**
14	67.71±12.73	58.95±27.94	90.24±5.64	73.43±21.20	94.42±13.17	**100.00±0.00**	90.50±6.49	91.58±4.46	90.23±5.18
15	98.89±1.26	63.08±16.90	94.25±4.12	96.6±1.12	82.50±21.85	97.75±4.67	91.50±3.28	97.66±2.40	**99.34±0.44**
OA	66.18±2.60	46.1±5.39	77.12±3.60	72.28±6.89	75.37±3.25	77.63±1.03	74.79±1.68	77.00±1.71	**82.02±0.53**
AA	68.52±2.34	46.18±7.24	81.29±2.11	75.83±6.91	78.50±3.57	79.83±1.00	77.33±1.14	79.51±1.62	**84.19±0.78**
Kappa	63.52±2.79	42.39±5.62	75.28±3.89	70.08±7.42	73.40±3.50	75.83±1.11	72.76±1.80	75.14±1.86	**80.57±0.58**

the large spectral variation of crop classes among green and near-infrared wavelengths, our proposed LAM can focus on important spectral features. For example, the classification accuracy using CF-Trans improved by 11.93% for soybean and 18.19% for corn over the DCFSL method. The SD and TD data are acquired by different sensors at different times and locations. Their data distributions and class are very different. Although, the KSC dataset are largely class different from the SD, the CF-Trans method achieved the best classification accuracy compared to other methods. It is shown that the CF-Trans method is effective in capturing interdomain invariant knowledge. The CF-Trans method works significantly in classifying samples with extreme imbalance and large class differences, such as the IP and KSC dataset. On the IP dataset, CF-Trans method achieved a 10.22% improvement in OA and a 11.46% improvement on Kappa versus the DCFSL method. High-spatial resolution remote-sensing images is beneficial to spatial feature extraction and improves classification performance. The UP, Salinas, and Houston university datasets all have better classification performance than the IP dataset. DLEM can effectively improve the classification of head classes, such as Salinas dataset, which achieved the best classification accuracy on Grapes_untrained and improved by 3.86% for the DCFSL method. On the IP dataset with extremely imbalanced samples, our method improves OA by more than 10% compared to state-of-the-art cross-domain methods. This indicates that DLEM effectively addresses the issue of imbalanced classification between head and tail classes. Additionally, the tail categories demonstrate excellent performance, with classification accuracies for Alfalfa and Oats in the IP dataset exceeding 95%. This demonstrates that CF-Trans enhances the classification efficacy of the head classes while concurrently preserving that of the tail classes. Based on the classification results, CF-Trans is not as effective as the global-based FGPA method for Grass-pasture-mowed and Hay-windrowed classes on IP dataset. Further analysis of the experimental data shows that Grass-pasture-mowed and Hay-windrowed classes are spread out across different geographic locations, highlighting that the global-based method is better at capturing overall context. The classification results of the KSC dataset indicate that the CF-Trans method achieves lower AA compared to HFSL, emphasizing the importance of capturing general features from natural images for effective classification. Despite certain limitations of the CF-Trans method, its overall classification performance was satisfactory.

D. Effects of Each Key Module

The CF-Trans network primarily encompasses LIDA, CRAN, and DLEM components. In this section, we will assess the impact of each key component of the CF-Trans network on the classification performance. The HU dataset was chosen for the ablation experiments due to its large-scale and significant differences in data style compared to other target domain datasets. The ablation experiments involve systematically removing individual key components from the entire framework to evaluate their impact.

As is shown in (a) in Table IX, the baseline method is based on FSL cross-domain HSIC method of DCFSL. In Section III, a detailed exposition of the CRAN is presented. (b) in Table IX displays the effectiveness of CRAN. The CRAN module

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TABLE IX
EACH KEY MODULE ABLATION EXPERIMENT OF CF-TRANS (FIVE LABELED SAMPLES FROM TARGET DOMAIN)

Method	CRAN	LAM	DLEM	OA(%)	Kappa(%)
(a) Baseline	-	-	-	77.54	75.72
(b) CF-Trans w/o LAM and DLEM	√	-	-	79.53	77.87
(c) CF-Trans w/o DLEM	√	√	-	80.67	79.12
(d) CF-Trans	√	√	√	**82.19**	**80.75**

is appended to CF-Trans, OA has improved from 77.54% to 79.53%, Kappa is improved from 75.72% to 77.87%. Experimental results illustrate that the CRAN module is capable of accurately extracting interdomain invariant features while preserving the intradomain original features.

The LAM is described in Section III. (c) In Table IX demonstrated the positive influence of this module on the classification accuracy. The LAM module is inserted into CF-Trans, OA has increased from 79.53% to 80.67%, and Kappa is increased from 77.87% to 79.12%. Experimental results show that LAM can capture more effective intradomain distinct knowledge with great domain differences.

The DLEM is clearly defined in Section III. (d) in Table IX, demonstrated by integrating this module into the framework, and it is clear that the classification effect has been noticeably enhanced. The DLEM module is joined to CF-Trans, OA is raised from 80.67% to 82.19%, and Kappa is raised from 79.12% to 80.75%. Experimental results reveal that the DLEM can effectively exploit the intradomain knowledge of the head classes.

E. Analysis of the Training Samples

To assess the superior performance of the CF-Trans method across varying numbers of training samples, a comparison of classification accuracies under various sample numbers is performed. The IP, UP, and Salinas datasets cover both agricultural and urban area data because these data exist with sample imbalance and feature similarities. These three datasets were selected for this experiment with different sample number. For fairness of the experiments, all experiments were executed on RTK 2080 card and with the same randomized seed. In this experiment, for each category, we train using 1–5 labeled samples separately, and then compared the OA, AA, and Kappa coefficient. As can be seen from Fig. 18(a)–(f), the classification accuracy gradually improves with the increase of classification samples, and CF-Trans achieves the best classification accuracy. The experimental results demonstrate that the CF-Trans method exhibits excellent classification performance for different sample sizes.

F. Computing Efficiency of LAM

To evaluate the computational burden of LAM, the parameters, training duration, and testing time are compared with three state-of-the-art approaches. Because of space limitations, we test the efficiency of the LAM here on IP data

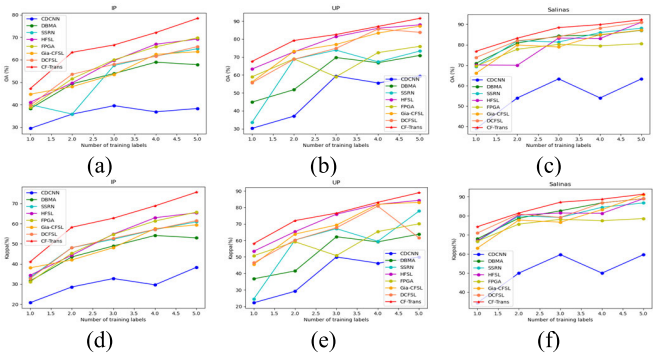


Fig. 18. Classification accuracies with different number of labeled samples on IP, UP, and Salinas datasets.

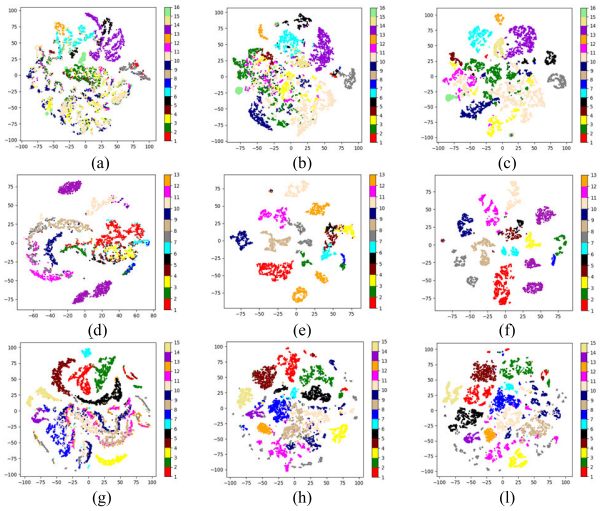


Fig. 19. IP, KSC, and HU datasets were visualized using 2-D t-SNE to analyze the features. The visualizations depict the data distributions of labeled samples in three feature spaces: the original (the first row), the Gia-CFSL (the second row), the CF-Trans (the three row). Each class is represented by a unique color. (a) IP Original features. (b) Gia-CFSL. (c) CF-Trans. (d) KSC original features. (e) Gia-CFSL. (f) CF-Trans. (g) HU original features. (h) Gia-CFSL. (l) CF-Trans.

that are difficult to classify and on HU data that have a large coverage. All experiments were conducted on the RTK5000 platform using the PyTorch framework, with a standardized 10 000 iterations. In Table X, LAM has less parameters and less training-testing time, and improves the classification accuracy of some very similar ground cover classes. From the perspective of the number of parameters, training time and testing time, CF-Trans does not have a significant increase in the number of parameters compared to the DCFSL method, but the accuracy is significantly improved. This demonstrates that CF-Trans achieves superior classification results with fewer model parameters and reduced training and testing times. Looking at the number of parameters, training time, and testing time, CF-Trans is significantly improved compared to HFSL and Gia-CFSL. This indicates that our method achieves superior classification results with fewer parameters compared to advanced cross-domain hyperspectral classification methods. CF-Trans requires extended training time when applied to the IP dataset, yet it exhibits substantial improvements with the Houston dataset, highlighting the advantageous efficacy of the LAM mechanism in large-scale mapping scenarios. Drawing from the experimental findings

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TABLE X
LAM COMPUTATIONAL BURDEN ANALYSIS

Datasets	Computing time	3DCNN	DBMA	SSRN	FGPA	HFSL	Gia-CFSL	DCFSL	DCFSL+CBAM	CF-Trans
IP	Params (M)	1.06	0.61	0.36	0.27	2.97	0.22	0.07	0.08	**0.07**
	Train time (s)	1000.00	26348.27	15000	319.63	5327.66	9025	2784.19	2833.64	**2717.87**
	Test time (s)	4.77	141.21	81.7	9.92	4.14	1.79	1.84	1.88	**1.77**
Houston	Params (M)	0.81	0.44	0.28	2.75	2.97	0.21	0.06	0.07	**0.07**
	Train time (s)	3000.00	24517.9	12000	4796.17	3238.35	5004.90	2552.35	2601.34	**2597.37**
	Test time (s)	17.12	166.55	46.00	7.20	5.05	2.48	2.56	2.63	**2.62**

that LAM effectively extracts intradomain spatial spectral knowledge in cases with domain differences and reduces computational burden.

G. Alignment Performance of CF-Trans

To evaluate the CF-Trans network, we visualize the features to analyze the feature alignment results. The IP dataset has a large gap with the SD data and a serious sample imbalance problem, with a large number of difficult to classify features. The KSC dataset has huge class differences to SD. The HU dataset covers a large area and better validates the robustness of our method. Because of the space limitation, we only analyze the effectiveness of our feature extraction on these three representative datasets. The original features of IP, KSC, and Houston are displayed in Fig. 19(a), (d), and (g). From Fig. 19(a), with a significant number of hybrid samples in categories 3 (Corn-mintill), 8 (Hay-windrowed), 10 (Soybean-notill) and 11 (Soybean-mintill) of IP data. Fig. 19(c), (f), and (l) shows CF-Trans network approach that makes intraclasses more similar and interclasses more distinguishable. This also proves the effectiveness of the CRAN, LAM, and DLEM modules.

V. CONCLUSION

In this article, a cross-domain HSIC framework is proposed to capture interdomain invariant knowledge and intradomain distinctiveness knowledge. To capture more discriminative intradomain spectral-spatial knowledge in scenarios with significant domain representation discrepancies, LAM is designed. It simultaneously considers information loss, accuracy, and efficiency. From the perspective of interdomain knowledge transformation to reduce the negative impact caused by inaccurate classifier predictions, the CRAN is proposed to capture interdomain invariant knowledge. The DLEM to balance the classification accuracy among tail and head classes is introduced. We conducted experiments in five different rural and urban areas, the CF-Trans method surpassed state-of-the-art HSIC approaches, particularly in scenarios with extremely imbalanced samples (e.g., the IP dataset in rural areas), significant category differences (e.g., the KSC dataset in rural–urban areas), and large-scale datasets (e.g., the Houston dataset in urban areas), exhibiting superior classification performance. The limited availability of head classes samples in FSL methods poses challenges in capturing the feature variations of head classes, resulting in a decline in head classes classification performance. In the future, we will conduct an in-depth study of this issue to enhance the application of the FSL method in the field of cross-domain HSIC.

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